

The Fractional Riccati–Müntz–Padé Theorem

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Abstract

For the fractional Riccati equation

$$D^\mu y = P(u) + Q(u) y + R(u) y^2 \quad (1)$$

with polynomial coefficients $P, Q, R \in \mathbb{C}[u]$, order $\mu \in (0, 1]$, and homogeneous Riemann–Liouville initial condition, the unique solution $y(t, u)$ is given explicitly by

$$y(t, u) = \lim_{M \rightarrow \infty} t^\mu \frac{N_M(t^\mu, u)}{D_M(t^\mu, u)} \quad (2)$$

where the rational pair (N_M, D_M) is produced by three steps: (i) compute $2M$ moments $a(0, u), \dots, a(2M-1, u)$ from the explicit quadratic recurrence (8); (ii) run Wheeler’s algorithm on those moments to obtain the Jacobi coefficients $\alpha(k), \beta(k)$ and orthogonal polynomials π_k ; (iii) set $D_M(z, u) = z^M \pi_M(z^{-1})$ and compute N_M by one Cauchy product. Total cost is $O(M^2)$ arithmetic operations, and convergence is Stahl-optimal: $|y - y_M|^{1/(2M)} \rightarrow e^{-g(z, K_A)} < 1$ in logarithmic capacity on compacta avoiding the Stahl compact $K_A(u)$, and uniformly on every $[0, T] \subset \mathbb{R}_+$ in the constant-coefficient case.

The denominator D_M is the characteristic polynomial of the truncated Jacobi operator J_M of the moment functional $\mathcal{L}[x^k] = a(k, u)$. Multiplication by x on $\text{span}\{\pi_0, \dots, \pi_{M-1}\}$ is a weighted shift with a one-dimensional cokernel at $\pi_M = D_M$; the resulting defect $J_M v(\lambda) = \lambda v(\lambda) - P_M(\lambda) e_M$ forces the numerator to degree exactly $M-1$. The $[M-1/M]$ type is not a bookkeeping choice but the codimension of the truncated Jacobi shift.

The proof rests on three ingredients. First, the Müntz–Tau recurrence produces the moments via the eigenvalue action of I^μ on monomials. Second, an auxiliary generating function F satisfies the closed-form algebraic quadratic $z R F^2 - (1 - z Q) F + P \Psi_\mu = 0$, whose branch structure controls the convergence domain, and where Ψ_μ is the Fox–Wright function

$$\Psi_\mu(z) = {}_1\Psi_1 \left[\begin{matrix} (1, \mu) \\ (1 + \mu, \mu) \end{matrix} ; z \right] = \sum_{k=0}^{\infty} \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)} z^k \quad (3)$$

Third, a Mellin–Barnes integral representation transfers meromorphic continuation from F to the solution generating function A via the Hadamard fixed-point equation $A = \Psi_\mu \odot B[A]$, and a self-consistent argument places K_A inside K_F . The Stahl convergence theorem for the generic two-valued algebraic class containing F and A is proved self-containedly in the appendix.

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1 The Approximant and the Algorithm

We begin with the explicit output of the method before any proofs. Given P, Q, R, μ , the following three-step procedure produces y_M .

1.1 Step 1: The Fox–Wright kernel and the moment recurrence

Definition 1. [Fox–Wright function ${}_1\Psi_1$] The Fox–Wright function is defined by

$${}_1\Psi_1 \left[\begin{matrix} (a, \alpha) \\ (b, \beta) \end{matrix} ; z \right] := \sum_{k=0}^{\infty} \frac{\Gamma(a + \alpha k)}{\Gamma(b + \beta k)} \frac{z^k}{k!} \quad (4)$$

convergent for all $z \in \mathbb{C}$ whenever $\beta - \alpha > -1$.

Definition 2. [Riccati–Fox–Wright kernel] The Riccati–Fox–Wright kernel of order $\mu \in (0, 1]$ is the specialization

$$\begin{aligned}\Psi_\mu(z) &:= {}_1\Psi_1\left[\begin{matrix}(1, \mu) \\ (1 + \mu, \mu)\end{matrix}; z\right] \\ &= \sum_{k=0}^{\infty} \psi_k z^k\end{aligned}\tag{5}$$

where

$$\psi_k := \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)}\tag{6}$$

The convergence condition $\beta - \alpha = \mu - \mu = 0 > -1$ is satisfied for all $\mu > 0$. The weights ψ_k are the eigenvalues of the Riemann–Liouville operator I^μ on the monomial basis $\{t^{(k+1)\mu}\}$ (Lemma 10 below); this eigenvalue action is the reason Ψ_μ appears rather than any other ${}_1\Psi_1$ specialization.

Remark 3. [Relation to the Mittag-Leffler function] At $\mu = 1$, $\psi_k = 1/(k+1)$ and $\Psi_1(z) = -(\log(1-z))/z$. For general μ , Ψ_μ interpolates between this logarithmic kernel at $\mu = 1$ and a delta-like concentration as $\mu \rightarrow 0^+$.

Theorem 4. [Müntz–Tau recurrence] The unique solution of (1) with homogeneous Riemann–Liouville initial condition satisfies $y(t, u) = \sum_{k \geq 0} a(k, u) t^{(k+1)\mu}$ on $[0, T_0]$ for some $T_0 > 0$, with

$$a(0, u) = P(u) \psi_0,\tag{7}$$

$$a(k, u) = \psi_k \left(Q(u) a(k-1, u) + R(u) \sum_{j+\ell=k-1} a(j, u) a(\ell, u) \right), \quad k \geq 1.\tag{8}$$

Each $a(k, u) \in \mathbb{C}[u][P, Q, R]$. Computing $a(0), \dots, a(2M-1)$ costs $O(M^2)$ operations.

Example 5. [$R=0$ oracle] When $R=0$ the recurrence linearizes to $a(k, u) = \psi_k Q^k P \psi_0$, giving the closed form

$$y(t, u) = \frac{P}{Q} (E_{\mu,1}(Q t^\mu) - 1),\tag{9}$$

where $E_{\mu,1}(z) = \sum_{k=0}^{\infty} z^k / \Gamma(k\mu + 1)$ is the Mittag-Leffler function. The $[M-1/M]$ Padé approximant of $A(z, u)$ must reproduce the rational approximant to $E_{\mu,1}(Qz)$ exactly for every M , providing a closed-form unit test for any numerical implementation.

1.2 Step 2: Wheeler’s algorithm

With moments $\mu(\ell) := a(\ell, u)$, quasi-definiteness (Definition 12) ensures the following recurrence is well-defined and produces the Jacobi coefficients.

Auxiliary array.

$$\sigma(-1, \ell) = 0 \quad (10)$$

$$\sigma(0, \ell) = \mu(\ell) \quad (11)$$

$$\sigma(k, \ell) = \sigma(k-1, \ell+1) - \alpha(k-1) \sigma(k-1, \ell) - \beta(k-1) \sigma(k-2, \ell) \quad (12)$$

Jacobi coefficients.

$$\alpha(0) = \mu(1) / \mu(0), \quad \beta(0) = \mu(0) \quad (13)$$

$$\alpha(k) = \frac{\sigma(k, k+1)}{\sigma(k, k)} - \frac{\sigma(k-1, k)}{\sigma(k-1, k-1)} \quad (14)$$

$$\beta(k) = \frac{\sigma(k, k)}{\sigma(k-1, k-1)} \quad (15)$$

Orthogonal polynomials.

$$\pi_{-1} = 0, \quad \pi_0 = 1 \quad (16)$$

$$\pi_{k+1} = (x - \alpha(k)) \pi_k - \beta(k) \pi_{k-1} \quad (17)$$

Cost: $O(M^2)$ operations.

1.3 Step 3: Assembling the approximant

Set

$$D_M(z, u) := z^M \pi_M(z^{-1}) \quad (18)$$

compute N_M by matching the first $2M$ coefficients of $A(z, u) = \sum_{k \geq 0} a(k, u) z^k$ against N_M/D_M (one Cauchy product, cost $O(M^2)$), and set

$$y_M(t, u) := t^\mu \frac{N_M(t^\mu, u)}{D_M(t^\mu, u)} \quad (19)$$

1.4 Operator-theoretic picture: why the type is $[M-1/M]$

The truncated Jacobi matrix

$$J_M := \begin{pmatrix} \alpha(0) & 1 & & & \\ \beta(1) & \alpha(1) & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \beta(M-1) & \alpha(M-1) & \end{pmatrix} \quad (20)$$

represents multiplication by x on $V_M := \text{span}\{\pi_0, \dots, \pi_{M-1}\}$. On the orthonormal basis $\hat{\pi}_k = \pi_k / \sqrt{\prod_{i=0}^k \beta(i)}$, this action is a weighted bidirectional shift:

$$\tilde{J} \hat{\pi}_k = \sqrt{\beta(k+1)} \hat{\pi}_{k+1} + \alpha(k) \hat{\pi}_k + \sqrt{\beta(k)} \hat{\pi}_{k-1} \quad (21)$$

Truncating to V_M creates a one-dimensional cokernel: the shift maps π_{M-1} toward $\pi_M = D_M$, which lies *outside* V_M . This missing component manifests as the defect

$$J_M v(\lambda) = \lambda v(\lambda) - P_M(\lambda) e_M \quad (22)$$

$$v(\lambda) := (P_0(\lambda), \dots, P_{M-1}(\lambda))^T$$

where e_M is the M -th standard basis vector. Hence $J_M v(\lambda) = \lambda v(\lambda)$ if and only if $P_M(\lambda) = 0$, i.e., λ is a zero of π_M , i.e., λ^{-1} is a zero of D_M . The resolvent at the $(0, 0)$ entry is

$$[(zI - J_M)^{-1}]_{0,0} = \frac{N_M(z)}{D_M(z)} \quad (23)$$

$$\deg N_M = M - 1 \quad (24)$$

$$\deg D_M = M \quad (25)$$

because the characteristic polynomial of the $M \times M$ matrix J_M is degree M , and its $(0, 0)$ -cofactor is degree $M - 1$. The $[M - 1 / M]$ type is therefore the codimension of the truncated Jacobi shift, not a convention.

2 Foundations

2.1 Riemann–Liouville calculus

Definition 6. [Riemann–Liouville operators] For $\mu \in (0, 1]$ and f locally integrable on $(0, \infty)$,

$$I^\mu f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s) ds \quad (26)$$

$$D^\mu f(t) := \frac{d}{dt}(I^{1-\mu} f)(t) \quad (27)$$

Definition 7. [Fractional Riccati equation]

$$D^\mu y(t, u) = P(u) + Q(u) y(t, u) + R(u) y(t, u)^2, \quad I^{1-\mu} y(0, u) = 0 \quad (28)$$

Lemma 8. [Volterra form] Equation (28) is equivalent to

$$y = I^\mu [P + Q y + R y^2] \quad (29)$$

Proof. Application of I^μ to (28), the composition formula [SKM, Theorem 2.4], and the homogeneous initial condition $I^{1-\mu} y(0, u) = 0$ kill the boundary term and yield (29). $\square$

Lemma 9. [Power rule] For $s > -1$,

$$I^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+\mu+1)} t^{s+\mu} \quad (30)$$

Proof. The substitution $\sigma = t\tau$ in (26) reduces the integral to $B(\mu, s+1) = \Gamma(\mu) \Gamma(s+1) / \Gamma(s+\mu+1)$. $\square$

Lemma 10. *[Eigenvalue action of I^μ] For $k \geq 0$,*

$$I^\mu t^{(k+1)\mu} = \psi_{k+1} t^{(k+2)\mu} \quad \psi_{k+1} = \frac{\Gamma((k+1)\mu+1)}{\Gamma((k+2)\mu+1)} \quad (31)$$

The monomials $\{t^{(k+1)\mu}\}_{k \geq 0}$ are eigenfunctions of I^μ and the Fox–Wright weights ψ_k of Definition 2 are the corresponding eigenvalues.

Proof. Apply Lemma 9 with $s = (k+1)\mu$. $\square$

Lemma 11. *[Radius of convergence of Ψ_μ] Ψ_μ has radius of convergence 1.*

Proof. By Stirling, $\psi_k \sim (k\mu)^{-\mu}$ as $k \rightarrow \infty$, so $\limsup_{k \rightarrow \infty} |\psi_k|^{1/k} = 1$. $\square$

2.2 Proof of the moment recurrence

The map $\mathcal{T}[y] := I^\mu(P + Qy + Ry^2)$ is a contraction on $C[0, T_0]$ for T_0^μ sufficiently small. Substituting the Müntz ansatz $y = \sum_{k \geq 0} a(k, u) t^{(k+1)\mu}$ and applying the eigenvalue action of Lemma 10 to match coefficients of $t^{(k+1)\mu}$ yields (7) at $k=0$ and (8) for $k \geq 1$.

2.3 Wheeler’s algorithm and the Padé connection

Definition 12. *[Quasi-definiteness] The sequence $\{a(k, u)\}$ is quasi-definite if the Hankel determinants $H_n(u) := \det(a(i+j, u))_{0 \leq i, j \leq n}$ satisfy $H_n(u) \neq 0$ for all $n \geq 0$.*

Theorem 13. *[Wheeler [Wheeler]] Under quasi-definiteness, $\sigma(k, k) \neq 0$ for all k , and the polynomials π_k produced by (17) satisfy*

$$\int \pi_j \pi_k d\lambda = \delta_{jk} \prod_{i=0}^k \beta(i) \quad (32)$$

Proof. Set $\tilde{\sigma}(k, \ell) := \int \pi_k x^\ell d\lambda$. This matches σ in initial data and recurrence, so they coincide. Induction shows $\pi_k \perp \{1, \dots, x^{k-1}\}$, giving $\sigma(k, \ell) = 0$ for $\ell < k$ and $\sigma(k, k) > 0$ by quasi-definiteness. The norm formula follows from $\langle \pi_k, \pi_k \rangle = \beta(k) \langle \pi_{k-1}, \pi_{k-1} \rangle$. $\square$

Theorem 14. *[Gragg–Padé connection [Gragg]] Under quasi-definiteness, $D_M(z) = z^M \pi_M(z^{-1})$ and the pair (N_M, D_M) defined in Step 3 is the $[M-1/M]$ Padé approximant of $A(z, u) = \sum_{k \geq 0} a(k, u) z^k$ at $z=0$.*

Theorem 15. *[Spectral identification] Under quasi-definiteness:*

1. The eigenvalues of J_M are exactly the zeros of π_M , hence the reciprocals of the zeros of $D_M(z) = z^M \pi_M(z^{-1})$.
2. For each eigenvalue λ of J_M , the corresponding right eigenvector is $v(\lambda) = (P_0(\lambda), \dots, P_{M-1}(\lambda))^T$ and the defect equation (22) holds.
3. As $M \rightarrow \infty$, the empirical eigenvalue measure $\nu_M := M^{-1} \sum_{\lambda \in \text{spec}(J_M)} \delta_\lambda$ converges weak-* to the equilibrium measure ν_{K_A} of the Stahl compact $K_A(u)$.

Proof. (1) By the determinantal identity $P_k(\lambda) = \det(\lambda I_k - \tilde{J}_k) / \prod_{i=2}^k \sqrt{\gamma_i}$, zeros of π_M coincide with $\text{spec}(J_M)$; the identification $D_M(z) = z^M \pi_M(z^{-1})$ gives the reciprocal relationship. (2) Direct substitution into J_M at rows $k < M-1$ yields $(J_M v(\lambda))_k = \lambda P_k(\lambda)$ by the three-term recurrence; at row $M-1$ the missing $P_M(\lambda)$ entry contributes $-P_M(\lambda) e_M$. (3) The zeros of D_M are reciprocals of $\text{spec}(J_M)$; Theorem 32 and (60) give weak-* convergence of the zero-counting measure to ν_{K_A} , and reciprocation is continuous on compacta avoiding the origin. $\square$

Remark 16. [Spectral measure: constant-coefficient case] When $P, Q, R \in \mathbb{C}$ are constants, A is meromorphic on $\mathbb{C}$ with simple poles at $\mathcal{P}_A$, and the spectral measure $\mu_A := \sum_{z^* \in \mathcal{P}_A} \text{Res}(A, z^*) \delta_{z^*}$ has moments $\mathcal{L}[x^k] = a(k)$. In this setting J_M is the truncated Jacobi operator of μ_A in the sense of Stone's theorem, and N_M/D_M is the rank- M Stieltjes transform of μ_A .

3 The Algebraic Quadratic and the Stahl Compact

Conventions. $[z^k] b := b_k$; $[z^k] (F \cdot G) = \sum_j f_j g_{k-j}$ (Cauchy product); $[z^k] (F \odot G) = f_k g_k$ (Hadamard product).

Definition 17. *[Auxiliary generating function] Set $f(0, u) := P\psi_0$ and for $k \geq 1$,*

$$f(k, u) := Q f(k-1, u) + R \sum_{j+\ell=k-1} f(j, u) f(\ell, u) + P\psi_k \quad (33)$$

Define $F(z, u) := \sum_{k \geq 0} f(k, u) z^k$

Remark 18. $F \neq A$: in (8) the bracket is premultiplied by ψ_k ; in (33) the source $P\psi_k$ enters additively. The bridge is the Hadamard fixed-point equation (38) of Section 4. At first order:

$$a(1, u) = \psi_1 (Q P\psi_0 + R (P\psi_0)^2) \quad (34)$$

$$f(1, u) = Q P\psi_0 + R (P\psi_0)^2 + P\psi_1 \quad (35)$$

which differ as elements of $\mathbb{C}[u]$ for generic P, Q, R, μ .

Theorem 19. *[Algebraic quadratic for F] F satisfies*

$$z R F^2 - (1 - z Q) F + P \Psi_\mu = 0 \quad (36)$$

with closed form

$$F(z, u) = \frac{(1 - z Q) - \sqrt{(1 - z Q)^2 - 4 z P R \Psi_\mu(z)}}{2 z R} \quad (37)$$

Proof. Let $G := z R F^2 - (1 - z Q) F + P \Psi_\mu$. One checks $[z^0] G = 0$ and $[z^k] G = 0$ for $k \geq 1$ by (33). Expanding the discriminant, $\sqrt{\Delta} = 1 - z(Q + 2 P R \psi_0) + O(z^2)$, shows the minus branch has constant term $P \psi_0 = f(0, u)$, the unique formal power-series root since $\partial_F [z R F^2 - (1 - z Q) F + P \Psi_\mu]|_{z=0} = -1$. $\square$

4 Meromorphic Continuation of A

4.1 Hadamard fixed-point equation

Lemma 20. *With $B[A] := P + z Q A + z R A^2$,*

$$A = \Psi_\mu \odot B[A] \quad (38)$$

Proof. $[z^0] B[A] = P$ gives $\psi_0 P = a(0)$. For $k \geq 1$, $[z^k] B[A] = Q a(k-1) + R \sum_{j+\ell=k-1} a(j) a(\ell)$, and multiplication by ψ_k recovers $a(k)$ by (8). $\square$

Remark 21. The Hadamard equation $A = \Psi_\mu \odot B[A]$ is not an independent construction: it is the coefficient-level restatement of the recurrence (8), written in generating-function language so that the Mellin–Barnes machinery below can act on it. Its sole role is to carry meromorphic continuation from F to A via the self-consistent fixed-point argument of Theorem 25.

4.2 Mellin transform of Ψ_μ

Lemma 22. *[Mellin transform] For $0 < \Re s < 1$,*

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \Gamma(1-s) \frac{\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)} =: M_{\Psi_\mu}(s) \quad (39)$$

M_{Ψ_μ} extends meromorphically to $\mathbb{C}$ with poles at $\mathbb{Z}_{\leq 0}$, $\mathbb{Z}_{\geq 1}$, and $(1+m)/\mu$ for $m \geq 0$. For $\mu \in (0, 1)$ the lattice $\{(1+m)/\mu\}$ survives uncanceled.

Proof. Write $\Psi_\mu(-x) = \sum_{k \geq 0} (-x)^k \varphi(k) / k!$ with $\varphi(k) = k! \psi_k$. The entire interpolation $\varphi(s) = \Gamma(s+1) \Gamma(s\mu+1) / \Gamma(s\mu+\mu+1)$ satisfies, by Stirling on a vertical line $\Re s = \sigma$,

$$|\varphi(\sigma + i\tau)| \sim C(\sigma, \mu) |\tau|^{\sigma + \mu\sigma - \mu + 1/2} e^{-(\pi/2)|\tau|} \quad (|\tau| \rightarrow \infty) \quad (40)$$

The exponential rate $\pi/2 < \pi$ satisfies Hardy's growth condition [Hardy], and Ramanujan's master theorem gives $\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \varphi(-s)$. The cancellation analysis between numerator and denominator poles is recorded in the statement. $\square$

4.3 Mellin–Barnes realisation

Lemma 23. Let $G(z, u) = \sum_{k \geq 0} g_k(u) z^k$ be holomorphic in $|z| < \rho$ with Ramanujan-normalised entire interpolation $\hat{G}_R(s, u) := \hat{G}(s, u) / \Gamma(s+1)$ satisfying the Gevrey-1 bound

$$|\hat{G}(\sigma + i\tau, u)| \leq K'_G C_G^{|\sigma|+1} |\Gamma(\sigma + i\tau + 1)| \quad (41)$$

Fix $c \in (0, 1)$ avoiding poles of M_{Ψ_μ} . For $|\arg(-z)| < \pi/2$ and $|z| < \min(1, \rho)$,

$$(\Psi_\mu \odot G)(z, u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s) \hat{G}_R(-s, u) (-z)^{-s} ds \quad (42)$$

and the integral continues meromorphically to $\tilde{X}$ minus the projected pole set of $M_{\Psi_\mu} \hat{G}_R$.

Proof. Closing the contour left, the residue of $\Gamma(s)$ at $s = -k$ is $(-1)^k / k!$; writing

$$M_{\Psi_\mu}(s) = \Gamma(s) H(s) \quad (43)$$

with

$$H(s) = \Gamma(1-s) \Gamma(1-s\mu) / \Gamma(\mu+1-s\mu) \quad (44)$$

one has

$$H(-k) = k! \psi_k \quad (45)$$

$$\hat{G}_R(k, u) = g_k(u) \quad (46)$$

$$(-z)^{-s}|_{s=-k} = (-1)^k z^k \quad (47)$$

, giving

$$\text{Res}_{s=-k}[\dots] = \psi_k g_k(u) z^k \quad (48)$$

. Summing over $k \geq 0$ recovers $\Psi_\mu \odot G$. Continuation to $\tilde{X}$ is by the identity principle. $\square$

4.4 Stahl compacts and meromorphic continuation

Lemma 24. [Stahl compact for F] There is a unique compact $K_F(u) \supseteq K_\Psi$ of minimal logarithmic capacity outside which $\sqrt{\Delta_F}$, hence F , is single-valued meromorphic on $\tilde{X}$, where $\Delta_F = (1-zQ)^2 - 4zPR\Psi_\mu$ and K_Ψ is the projected pole set of M_{Ψ_μ} .

Theorem 25. *[Meromorphic continuation of A] Under quasi-definiteness, A is meromorphic on $\tilde{X} \setminus K_A(u)$ with $K_A(u) \subseteq K_F(u)$.*

Proof. Step 1 (Gevrey-1 bound). Set $C_0 = \psi_0 = 1/\Gamma(\mu + 1)$, $M_P = |P|$, $M_Q = |Q|$, $M_R = |R|$,

$$K_G := C_0 \max(M_P, 1) \quad (49)$$

$$C_G := C_0 (M_Q + 2 K_G M_R) + 1 \quad (50)$$

Induction using

$$\sum_{j+\ell=n} j! \ell! \leq (n+1)! \quad (51)$$

and

$$\frac{C_0 (M_Q + K_G M_R)}{C_G} < 1 \quad (52)$$

(by construction of C_G) establishes

$$|a(k, u)| \leq K_G C_G^k k! \quad (53)$$

; a parallel bound

$$|g_k(u)| \leq K_G'' C_G^{k+1} k! \quad (54)$$

follows from the defining recurrence of $f(k)$.

Step 2 (Cauchy interpolation). The Cauchy integral

$$\hat{G}(s, u) := \oint_{|z|=r} G(z, u) z^{-s-1} \Gamma(s+1) dz / (2\pi i) \quad (55)$$

satisfies the Gevrey-1 bound (41) via the Phragmén–Lindelöf principle in each vertical strip. Carlson’s theorem guarantees uniqueness as the entire interpolation of $\{k! g_k\}_{k \geq 0}$ of exponential type $< \pi$.

Step 3 (Mellin–Barnes). Lemma 23 applies to $G = B[A]$ and yields the meromorphic continuation of $A = \Psi_\mu \odot G$ to $\tilde{X} \setminus (K_\Psi \cup S_G)$.

Step 4 (Self-consistent fixed point). The map $\Phi: A \mapsto \Psi_\mu \odot B[A]$ sends the space $\mathcal{M}(K)$ of formal series with meromorphic continuation on $\tilde{X} \setminus K$ to itself. The difference $A - F$ satisfies a linear inhomogeneous functional equation with forcing meromorphic on $\tilde{X} \setminus K_F$ (Lemma 26), so $A \in \mathcal{M}(K_F)$. The minimal compact $K_A \subseteq K_F$ exists by Theorem 31. $\square$

Lemma 26. *[Branch transfer] The branch locus of A is contained in the branch locus of F together with K_Ψ .*

Proof. Subtracting the $f(k)$ recurrence (33) from the $a(k)$ recurrence (8) gives a linear inhomogeneous functional equation for $A - F$ in $\mathcal{K}(\Psi_\mu)$ with forcing meromorphic on $\tilde{X} \setminus K_F$. The analytic implicit function theorem applies wherever $2zRF - (1 - zQ) \neq 0$, whose zero set lies in K_F , giving a unique meromorphic $A - F$ on $\tilde{X} \setminus K_F$. $\square$

5 The Fractional Riccati–Müntz–Padé Theorem

Theorem 27. *[Fractional Riccati–Müntz–Padé theorem] Under quasi-definiteness, the approximant $y_M(t, u) = t^\mu N_M(t^\mu, u) / D_M(t^\mu, u)$ satisfies*

$$y(t, u) = \lim_{M \rightarrow \infty} y_M(t, u) \quad (56)$$

in logarithmic capacity on compact subsets of the t -universal-cover preimage of $\tilde{X} \setminus K_A(u)$.

Proof. $y(t, u) = t^\mu A(t^\mu, u)$ by Theorem 4. The denominator sequences for both $[M-1/M]$ and $[M/M]$ approximants of A are identical (both are $\{\pi_M\}$); Theorem 32 gives $N_M/D_M \rightarrow A$ in capacity on compacta of $\tilde{X} \setminus K_A$. The map $t \mapsto t^\mu$ is a C^1 diffeomorphism on the principal-branch sector with capacity-transformation identity $\text{cap}(\{t: t^\mu \in K\}) = \text{cap}(K)^{1/\mu}$ (Ransford [Ransford, Thm. 5.2.5]), transferring convergence sheetwise to the t -plane. $\square$

Lemma 28. *[No real poles] In the constant-coefficient case $P, Q, R \in \mathbb{C}$, $R \neq 0$, the generating function $A(z)$ has no poles on $(0, \infty)$.*

Proof. Each pole z^* of A is simple with residue $\text{Res}(A, z^*) = -(z^*)^{(\mu-1)/\mu} / (R\mu)$. For $z^* \in (0, \infty)$ this is nonzero, but the Volterra fixed point is holomorphic on any compact real interval where it converges, giving a contradiction. Hence no real poles exist. $\square$

Corollary 29. *[Uniform convergence on $\mathbb{R}_+$] In the constant-coefficient case, for every compact $[0, T] \subset \mathbb{R}_+$,*

$$\sup_{t \in [0, T]} |y(t) - y_M(t)| \longrightarrow 0 \quad \text{as } M \rightarrow \infty. \quad (57)$$

Proof. By Lemma 28, $\delta := \text{dist}([0, T], \mathcal{P}_A) > 0$. On $U_\delta := \{z \in \mathbb{C} : \text{dist}(z, [0, T]) < \delta/2\}$, A is holomorphic and bounded. Convergence in capacity on U_δ (Theorem 32) implies uniform convergence on $[0, T]$: the capacity-zero exceptional sets cannot intersect $[0, T]$ for all sufficiently large M , and a normal-families bound finishes the argument. The substitution $z = t^\mu$ transfers uniform convergence from $[0, T^\mu]$ to $[0, T]$. $\square$

6 A Direct Proof of Stahl's Theorem for the Generic Two-Valued Class over $\mathbb{C}(z, \Psi_\mu)$

This appendix proves Stahl's extremal-domain and convergence-in-capacity theorems for the class $\mathcal{A}$ of functions Φ on $\tilde{X}$ (the universal cover of $\mathbb{C} \setminus \{0\}$ minus a finite set) satisfying: Φ is two-valued algebraic over $\mathcal{K} := \mathbb{C}(u)(z, \Psi_\mu(z))$, generically irreducible; the discriminant $\Delta_\Phi \in \mathcal{K}$ is meromorphic on $\tilde{X} \setminus K_\Psi$ with isolated zeros and poles; Φ is holomorphic at $z=0$. Both F from (37) and A from Theorem 25 belong to $\mathcal{A}$.

6.1 Logarithmic capacity

$$I[\nu] := \iint \log \frac{1}{|z - w|} d\nu(z) d\nu(w) \quad (58)$$

$$\text{cap}(K) := e^{-\inf_{\nu} I[\nu]} \quad (59)$$

The infimum is attained by a unique equilibrium measure ν_K when $\text{cap}(K) > 0$.

Lemma 30. *[Monotonicity]* $\text{cap}(K_1 \cup K_2) \geq \max(\text{cap}(K_1), \text{cap}(K_2))$.

6.2 Extremal domain

Theorem 31. *[Extremal domain]* Let $\Phi \in \mathcal{A}$ with branch locus contained in $K_0 \supseteq K_\Psi$. There exists a unique compact K_Φ with $K_\Psi \subseteq K_\Phi \subseteq K_0$ of minimal logarithmic capacity outside which Φ is single-valued meromorphic on $\tilde{X}$.

Proof. Let $\mathcal{S}$ be the family of admissible compacts. $\mathcal{S}$ is nonempty and closed under directed intersections by the identity principle. The capacity functional is upper semicontinuous; Zorn's lemma produces a minimal element K_Φ . Uniqueness: if K_Φ and K'_Φ both minimize capacity, their intersection lies in $\mathcal{S}$ and the identity principle glues the two single-valued continuations, giving $\text{cap}(K_\Phi \cap K'_\Phi) \leq \min(\text{cap}(K_\Phi), \text{cap}(K'_\Phi))$ with equality forced by minimality; Stahl's $\mathcal{S}$ -property then yields $K_\Phi = K'_\Phi$. $\square$

6.3 Convergence in capacity

Theorem 32. *[Convergence in capacity]* Let $\Phi \in \mathcal{A}$ with extremal compact K_Φ and diagonal Padé sequence $\{N_M/D_M\}$ built from the Taylor series of Φ at $z = 0$. Then $N_M/D_M \rightarrow \Phi$ in logarithmic capacity on every compact subset of $\tilde{X} \setminus K_\Phi$.

Proof. The zero-counting measure

$$\nu_M := M^{-1} \sum_{D_M(z)=0} \delta_z \quad (60)$$

converges weak- $*$ to the equilibrium measure ν_{K_Φ} .

Step 1 (Hermite's formula).

$$\Phi - R_M = \frac{\Pi_M(z)}{D_M(z)^2} \cdot \frac{1}{2\pi i} \oint_{\gamma} \frac{D_M(\zeta)^2 \Phi(\zeta)}{\Pi_M(\zeta)(\zeta - z)} d\zeta,$$

where γ encircles K_Φ and Π_M is a monic polynomial of degree M unrelated to the orthogonal polynomials P_M .

Step 2 (Capacity estimate). $|D_M(z)|^{1/M} \rightarrow e^{g(z, K_\Phi)} \text{cap}(K_\Phi)$ in capacity (Stahl's main lemma [StahlExtremal, StahlExtremalII]).

Step 3 (Vanishing in capacity). $|\Phi(z) - R_M(z)|^{1/(2M)} \rightarrow e^{-g(z, K_\Phi)} < 1$ off K_Φ , giving exponential convergence in capacity. $\square$

6.4 Absence of Froissart doublets on $\mathbb{R}_+$

Froissart doublets are near-cancelling pole–zero pairs of Padé approximants with no counterpart in the approximated function. They do not arise here because the residues of A at its poles are nonzero and computable, preventing any silent cancellation on $\mathbb{R}_+$.

Lemma 33. *[No Froissart doublets] In the constant-coefficient case, no pole of any D_M on $\mathbb{R}_+$ can cancel silently against a zero of N_M on $\mathbb{R}_+$.*

Proof. By Lemma 28, A has no poles on $(0, \infty)$. A Froissart doublet at $z^* \in (0, \infty)$ would require a pole of D_M at z^* paired with a zero of N_M at z^* ; but the residue formula $\text{Res}(A, z^*) = -(z^*)^{(\mu-1)/\mu} / (R\mu) \neq 0$ means any such pole in the approximant is genuine and cannot be zeroed out in the numerator consistently with the moment-matching condition. $\square$

Bibliography

- [SKM] S. G. Samko, A. A. Kilbas, O. I. Marichev, *Fractional Integrals and Derivatives*, Gordon and Breach, 1993.
- [StahlExtremal] H. Stahl, *Extremal domains associated with an analytic function I*, Constr. Approx. **1** (1985), 89–110.
- [StahlExtremalII] H. Stahl, *Extremal domains II*, Constr. Approx. **1** (1985), 111–137.
- [StahlConvergence] H. Stahl, *Convergence of Padé approximants to functions with branch points (preliminary)*, Constr. Approx. **4** (1988), 631–642.
- [StahlBranch] H. Stahl, *Convergence of Padé approximants to functions with branch points*, J. Approx. Theory **91** (1997), 139–204.
- [Wheeler] J. C. Wheeler, *Modified moments and Gaussian quadratures*, Rocky Mountain J. Math. **4** (1974), 287–296.
- [Gragg] W. B. Gragg, *The Padé table and its relation to certain algorithms of numerical analysis*, SIAM Review **14** (1972), 1–62.
- [Hardy] G. H. Hardy, *Ramanujan: Twelve Lectures on Subjects Suggested by His Life and Work*, Cambridge University Press, 1940; reprinted Chelsea, New York, 1959.
- [Ransford] T. Ransford, *Potential Theory in the Complex Plane*, London Math. Soc. Student Texts **28**, Cambridge Univ. Press, 1995.